

# Hashing — Exercises 11.2

## Collision Resolution by Chaining

### Exercise 11.2-1

Suppose we use a hash function  $h$  to hash  $n$  distinct keys into an array  $T$  of length  $m$ . Assuming simple uniform hashing, what is the expected number of collisions? More precisely, what is the expected cardinality of

$$\{\{k, l\} : k \neq l \text{ and } h(k) = h(l)\}?$$

### Solution

For each pair of keys  $k, l$  with  $k \neq l$ , define

$$X_{kl} = \begin{cases} 1 & \text{if } h(k) = h(l) \\ 0 & \text{otherwise} \end{cases}$$

Under simple uniform hashing:

$$\Pr(h(k) = h(l)) = \frac{1}{m}$$

Thus:

$$E[X_{kl}] = \frac{1}{m}$$

Let

$$Y = \sum_{k < l} X_{kl}$$

be the total number of collisions.

By linearity of expectation:

$$E[Y] = \sum_{k < l} E[X_{kl}] = \binom{n}{2} \cdot \frac{1}{m} = \frac{n(n-1)}{2m}$$

$$\boxed{E[\text{collisions}] = \frac{n(n-1)}{2m}}$$

### Exercise 11.2-2

Insert the keys 5, 28, 19, 15, 20, 33, 12, 17, 10 into a hash table with 9 slots using

$$h(k) = k \bmod 9.$$

### Solution

Compute:

$$\begin{aligned}5 \bmod 9 &= 5 \\28 \bmod 9 &= 1 \\19 \bmod 9 &= 1 \\15 \bmod 9 &= 6 \\20 \bmod 9 &= 2 \\33 \bmod 9 &= 6 \\12 \bmod 9 &= 3 \\17 \bmod 9 &= 8 \\10 \bmod 9 &= 1\end{aligned}$$

Using insertion at head:

| Slot | Chain                              |
|------|------------------------------------|
| 0    | <i>NIL</i>                         |
| 1    | $10 \rightarrow 19 \rightarrow 28$ |
| 2    | 20                                 |
| 3    | 12                                 |
| 4    | <i>NIL</i>                         |
| 5    | 5                                  |
| 6    | $33 \rightarrow 15$                |
| 7    | <i>NIL</i>                         |
| 8    | 17                                 |

### Exercise 11.2-3

Professor Marley keeps each chain sorted. How does this affect running times?

### Solution

Let average chain length be  $\alpha$ .

- Successful search:  $\Theta(1 + \alpha)$  (unchanged)
- Unsuccessful search:  $\Theta(1 + \alpha)$  (may terminate early but same asymptotic)
- Insertion:  $\Theta(1 + \alpha)$  (must find correct position)
- Deletion:  $\Theta(1 + \alpha)$  if key given;  $O(1)$  if pointer given

Sorting does not change asymptotic search time, but increases insertion cost.

### Exercise 11.2-4

Suggest how to allocate/deallocate storage using a free list. Does it need to be doubly linked?

### Solution

Each slot stores:

- A flag (free/used)
- Either (element + pointer)
- Or two pointers (for free list)

Free slots are linked in a free list.

Operations:

**Insertion:**

- Remove a slot from free list
- Place element

**Deletion:**

- Remove element
- Return slot to free list

All operations take expected  $O(1)$  time.

The free list should be **doubly linked** if arbitrary deletions are needed. Otherwise, singly linked suffices if removing only from head.

### Exercise 11.2-5

If  $|U| > nm$ , show that  $U$  contains  $n$  keys that hash to the same slot.

### Solution

There are  $m$  slots.

Suppose each of  $m - 1$  slots contains at most  $n - 1$  elements.

Then remaining slot contains at least:

$$|U| - (m - 1)(n - 1)$$

Since  $|U| > nm$ :

$$> nm - (m - 1)(n - 1) = n + m - 1 \geq n$$

Thus one slot contains at least  $n$  keys.

Worst-case search time:

$$\boxed{\Theta(n)}$$

### Exercise 11.2-6

Select a key uniformly at random in expected time

$$O(L(1 + 1/\alpha))$$

where  $L$  is longest chain.

### Solution

View table as  $m$  rows, each row length  $\leq L$ .

Pick a random slot and a random position.

Probability of selecting a real key:

$$\frac{n}{mL}$$

Expected trials:

$$\frac{mL}{n} = \frac{L}{\alpha}$$

Each trial costs  $O(1)$ . Final chain scan costs  $O(L)$ .

Total expected time:

$$\boxed{O(L(1 + 1/\alpha))}$$